

PAPER

nf-core/viralmetagenome: A Novel Pipeline for Untargeted Viral Genome Reconstruction

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Abstract

Motivation: ~~Eukaryotic viruses present significant challenges for genome reconstruction and variant analysis~~ Reconstructing eukaryotic viral genomes from metagenomic data is challenging due to their extensive diversity and potential genome segmentation. ~~While de novo assembly followed by Current approaches often rely on labor-intensive manual curation for reference database matching selection and scaffolding is a commonly used approach, the manual execution of this workflow is extremely time-consuming, particularly due to the extensive reference curation required limiting scalability for large studies or rapid outbreak response. Here, we~~ We address the critical need for an automated, scalable pipeline ~~that can efficiently handle~~ for efficient viral metagenomic analysis without manual intervention.

Results: We present nf-core/viralmetagenome, a comprehensive ~~viral metagenomic~~ Nextflow pipeline for ~~the~~ untargeted ~~genome reconstruction and variant analysis of eukaryotic DNA and RNA viruses~~. ~~Viralmetagenome is implemented as a Nextflow workflow that processes from short-read metagenomic or hybridisation capture enriched samples to automatically detect and assemble viral genomes, while also performing variant analysis.~~ The pipeline ~~features automates the entire process from read preprocessing to consensus generation, integrating multiple de novo assemblers, automated reference selection, and iterative consensus refinement. It features robust quality control metrics, comprehensive extensive documentation, and seamless integration with containerization technologies, including portability via Docker and Singularity. We demonstrate validated the utility and accuracy of our approach through validation pipeline on both diverse simulated and real datasets, showing robust performance across diverse viral families in demonstrating its ability to recover high-quality genomes from complex metagenomic samples and resolve co-infections, making it a powerful tool for viral surveillance.~~

Availability: nf-core/viralmetagenome is freely available at <https://github.com/nf-core/viralmetagenome> with comprehensive documentation at <https://nf-co.re/viralmetagenome>. Archival code repository snapshots are published at zenodo with doi: <https://doi.org/10.5281/zenodo.17524074>.

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Supplementary information: Supplementary data are available at <https://github.com/Joon-Klaps/nf-core-viralmetagenome-manuscript> online.

Key words: viralmetagenome, bioinformatic pipeline, nextflow, viral metagenomics, viral assembly, viral variant analysis

Introduction

Reconstructing viral genomes from metagenomic sequencing data presents considerable computational challenges, particularly for viruses exhibiting extensive genetic diversity (Baaijens et al. 2017, Meleshko et al. 2021). This ~~diversity is further compounded by segmented genomes in families like challenge~~ is amplified in segmented viruses such as

influenza, rotavirus, and bunyaviruses, where individual segments can evolve under distinct selective pressures and ~~reassort, contributing to a complex landscape for genome reassortment complicates reconstruction. While pipelines are often designed to some pipelines target specific viruses and their subtypes (Shepard et al. 2016), accurate and complete genome reconstruction of genome reconstruction from samples with unknown references typically requires viral strains typically~~

requires time-consuming manual curation of contigs and reference matching (de Vries et al. 2021). This manual curation process is time-consuming, making it impractical for large-scale metagenome studies or rapid response scenarios that involve emerging viral outbreaks of unknown origin/outbreak responses impractical.

To address these limitations, we developed nf-core/viralmetagenome, a comprehensive pipeline specifically designed for untargeted viral genome reconstruction using short read sequences. The pipeline is developed using Nextflow (Di Tommaso et al. 2017) within the nf-core framework (Ewels et al. 2020) (Ewels et al. 2020, Langer et al. 2025), ensuring reproducibility through containerization with Docker (Merkel 2014) and Singularity (Kurtzer et al. 2017), and enabling portability across computational platforms such as local desktops, high-performance clusters and cloud environments.

Pipeline Description

nf-core/viralmetagenome implements an automated workflow performing for *de novo* assembly, reference matching, and iterative consensus refinement for the reconstruction of viral genomes without prior target knowledge. The pipeline consists of comprises five major stages: read preprocessing, metagenomic diversity assessment, contig assembly and scaffolding, iterative consensus refinement with variant analysis, and quality control (Figure 1). While this manuscript highlights key differences between particular tools, unless otherwise specified, the pipeline offers multiple options to accommodate established user workflows and preferences. In depth tool details are available. Detailed tool descriptions are provided in Supplementary Table 1.

Read preprocessing

~~Input reads~~ Input reads are provided via sample sheets containing sample names and paths to short-read FASTQ paths/files. These are preprocessed using FastQC and have their adapters trimmed with trimmed for adapters using fastp (Chen et al. 2018) (default) or Trimmomatic (Bolger et al. 2014). Fastp is overall faster and has offers automated adapter detection and trimming faster processing (Chen et al. 2018). UMI deduplication is implemented using HUMID (Laros 2024) and once reads are mapped to a reference with for raw reads, or UMI-tools after mapping (Smith et al. 2017). Multiple sequencing runs are can be merged after trimming by specifying merge group identifiers in the sample sheet. Complexity Optional filtering with bbduk (Bushnell 2014) or PRINSEQ++ (Cantu et al. 2019) removes low-complexity sequences containing repetitive elements where, while host reads are removed with Kraken2 (Wood et al. 2019).

Metagenomic diversity assessment

Taxonomic classification of preprocessed reads is performed using two complementary approaches — Kaiju approaches — Kaiju (Menzel et al. 2016) and Kraken2 (Wood et al. 2019) — to — to maximise detection sensitivity across diverse viral families. Results from both classifiers are visualised using Krona (Ondov et al. 2011).

De novo De novo assembly and clustering

The assembly workflow implements employs a multi-assembler approaches followed by clustering and scaffolding approach

where contigs generated by all tools are pooled and processed jointly. *De novo* assembly is performed using SPAdes (Meleshko et al. 2021) (RNAviral mode), MEGAHIT (Li et al. 2016), and Trinity (Grabherr et al. 2011), capitalizing on their distinct algorithmic strengths to maximise genome recovery across diverse viral families and variable read depths.

~~Reference identification uses BLASTn (Altschul et al. 1990)~~ Subsequently, a BLASTn (Altschul et al. 1990) search against the Reference Viral Database (RVDB) (Goodacre et al. 2018) , retaining top five hits to facilitate identification of related genomic segments and appropriate reference sequences for identifying candidate references for each contig. All contigs and their respective references are then subjected to clustering. This strategy enables contig scaffolding against the cluster centroid — whether an external reference or a *de novo* contig scaffolding and clustering — thereby combining outputs from multiple assemblers into unified scaffolds.

Clustering is performed in two sequential stages. First, taxonomic pre-clustering groups contigs based on taxonomic classification using Kraken2 (Wood et al. 2019) and Kaiju (Menzel et al. 2016), with optional filtering to focus on specific taxonomic clades for more targeted analyses. Second, nucleotide similarity clustering within taxonomic groups is performed using CD-HIT-EST (Li and Godzik 2006) (default), VSEARCH (Rognes et al. 2016), MMseqs2 (Steinegger and Söding 2017), vRhyme (Kieft et al. 2022), or Mash (Ondov et al. 2019). All tools are valid options, though performance may vary depending on the dataset (Zielezinski et al. 2025, Steinegger and Söding 2017).

~~As an optional filtering step of contig clusters, after~~ Optionally, following assembly and extension, reads can be mapped to all contigs using BWA-MEM2 (Vasimuddin et al. 2019) (default), BWA (Li 2013), or bowtie2 or Bowtie2 (Langmead et al. 2019). Clusters are filtered based on the total percentage of reads mapping to all contigs within a cluster, allowing identification and removal of clusters that likely represent assembly artefacts resulting from low read coverage.

enabling the removal of assembly artefacts. For the final scaffolding step, all cluster members are mapped to the cluster representative or centroid using Minimap2 (Li 2018), followed by consensus calling with iVar (Grubaugh et al. 2019) to generate reference-assisted assemblies. Regions with zero coverage depth can optionally be represented by the reference genome to produce a more complete scaffold genome for consensus calling scaffolds.

Iterative consensus refinement Consensus generation and variant calling

The consensus module supports external reference-based analysis and scaffold refinement. Users can provide a separate reference genome or reference set for each sample with — mapping constraints; when a reference specific reference genomes; if a set is provided, Mash (Ondov et al. 2019) selects the most similar can be selected using Mash (Ondov et al. 2019) one. The selected reference genome undergoes a single round of refinement to generate the consensus.

~~Scaffold refinement performs~~ In contrast, scaffold refinement employs an iterative approach, performing up to 4 iterative cycles (default 2) . Each iteration maps to polish the consensus sequence. Both processes map reads using BWA-MEM2 (Vasimuddin et al. 2019) (default), or

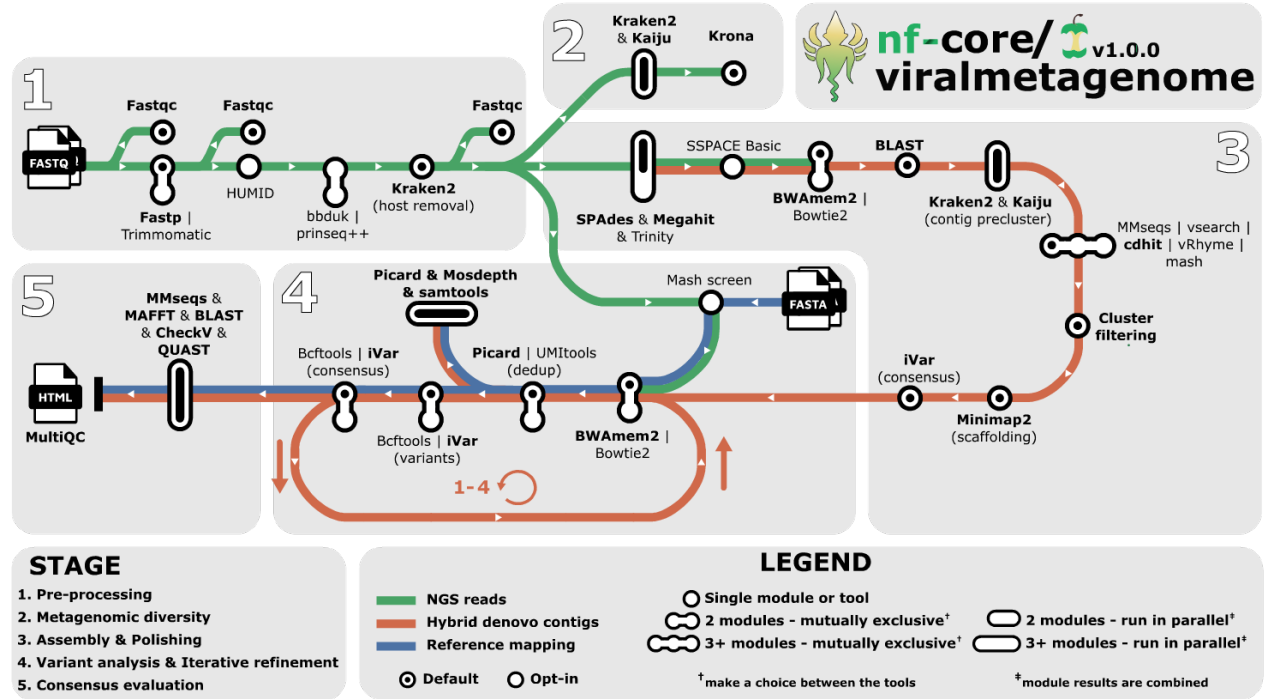


Fig. 1. Visual overview of the Overview of the nf-core/viralmetagenome pipeline for untargeted viral genome reconstruction. Default options are highlighted in bold. nf-core/viralmetagenome pipeline for untargeted viral genome reconstruction processes short-read data through pre-processing, metagenomic diversity assessment, *de novo* assembly with multiple assemblers, scaffolding with automated reference identification and **contig**-taxonomy-guided clustering of **contigs**, and iterative consensus refinement through read mapping and **variant-consensus** calling. Quality control metrics, assembly statistics, and coverage data are integrated into interactive MultiQC reports and standardised overview tables for downstream analysis.

Bowtie2 (Langmead et al. 2019), BWA (Li 2013), or bowtie2 (Langmead et al. 2019) to the consensus, followed by variant calling with BCFtools (Danecek et al. 2021) or iVar (Grubaugh et al. 2019). Benchmarking by Bassano et al. (Bassano et al. 2022) showed that BCFtools outperformed (default). Benchmarking (Bassano et al. 2022) indicated that BCFtools outperforms iVar in precision and recall, where while iVar identified more low-frequency-low-frequency variants. Users are recommended to consider prioritising sensitivity or specificity when selecting the variant caller. Final variant calling is performed on the polished consensus to characterise single nucleotide variants.

Consensus Quality Control

Quality control employs CheckV (Nayfach et al. 2021) for completeness estimation to assess completeness, BLASTn (Altschul et al. 1990) for reference similarity, and MMseqs2 (Steinegger and Söding 2017) against the annotated database Virosaurus (Gleizes et al. 2020). These analyses enable species identification, genomic segment classification, host prediction, and any other retrieval of additional metadata embedded within the reference databases.

The refinement progression consensus refinement is evaluated through sequence alignment with MAFFT (Katoh et al. 2002), which compares final consensus genomes against *de novo* contigs, intermediate consensus sequences from iterative cycles, and the scaffolding reference. All tool metrics are compiled into an interactive MultiQC report (Ewels et al. 2016). Additionally, key metrics are extracted from the

MultiQC report and compiled into standalone overview tables to facilitate downstream analysis across all processed samples.

Applications

To assess the performance of nf-core/viralmetagenome under challenging scenarios, we simulated coinfection scenarios by mixing paired-end reads from public HIV-1 genomes with varying diversity (80-99%), resulting in 13 samples (See supplementary table 2). nf-core/viralmetagenome successfully identified coinfections in all mixed samples when genetic similarity was low to moderate ($\leq 96.7\%$).

We validated nf-core/viralmetagenome performance on We validated the pipeline's performance using a diverse set of real metagenomic samples spanning containing human and plant pathogens. Here, the pipeline successfully generated nf-core/viralmetagenome successfully reconstructed high-quality or near-complete genomes across viral families including segmented consensus genomes across various viral families, including segmented viruses (Lassa virus, Orthonairovirus, Orthonairovirus, Tomato spotted wilt tospovirus) and non-segmented viruses (SARS-CoV-2, West Nile virus, Potato virus Y, Youcai mosaic virus, and Monkeypox virus).

Processing; Supplementary methods S1). The pipeline proved efficient, processing 28 samples (supplementary methods) required in 412 CPU hours and a maximum with a maximum memory footprint of 79GB RAM on an HPC, excluding taxonomic classification steps. The automated reference selection offers substantial improvements over

manual curation by reducing processing time while preserving reconstruction accuracy. Performance correlates strongly with reference database comprehensiveness, as consensus genomes tended to be more complete and similar to the true consensus sequence when the scaffolding reference was closer to the true viral genome on a high-performance cluster. To facilitate local deployment, we implemented a `-profile local` option, which utilizes a smaller viral RefSeq database instead of the full RVDB database for Kaiju, reducing the pipeline's peak RAM usage to approximately 18GB.

To evaluate the pipeline's robustness in complex scenarios, we conducted two targeted experiments. First, we assessed its ability to resolve co-infections using a simulated mixture of viral genomes (Supplementary Methods S2). The pipeline successfully distinguished and reconstructed distinct co-infecting strains, provided the genetic divergence was sufficient ($\leq 88.7\%$ ANI; Supplementary Figure 1).

The second part of our investigation explored the critical role of the scaffolding reference in consensus genome reconstruction. This emphasizes We selected sequencing data from seven Lassa virus-positive human clinical samples with diverse viral loads (Supplementary Methods S3; Supplementary Figure 2). For these seven samples, a significantly higher sequencing depth was observed for the S segment than for the L segment (Supplementary Figure 3). To establish a benchmark, a baseline consensus genome was first generated for each sample using automated reference selection from the full unclustered-RVDB (U-RVDB) reference pool. We then systematically challenged the scaffolding process by providing it a reference genome from a range of references that have varying degrees of similarity to the baseline consensus (64%–100% ANI; Supplementary Figure 4).

Comparing these generated consensus sequences against their respective baseline highlights the utility of reference scaffolding (Figure 2), particularly in low read depth scenarios when *de novo* assembly was challenging. For the high-coverage S segment, *de novo* assembly was consistently sufficient to reconstruct the full genome, rendering the choice of scaffolding reference largely redundant (Figure 2A-B). In contrast, the low-coverage L segment proved more challenging; *de novo* assembly produced shorter, incomplete fragments, demonstrating that the pipeline's ability to identify and utilize a closely related external reference was essential to bridge gaps and recover a complete segment consensus genome (Figure 2C-D).

These findings emphasise the need to keep databases like RVDB (Goodacre et al. 2018) and *Virosaurus* (Gleizes et al. 2020) up-to-date. Since nf-core/viralmetagenome is primarily designed for eukaryotic viruses, bacteriophage analysis requires different approaches and users interested in bacteriophage analysis are encouraged to explore pipelines targeting designed to target phages such as VIRify (Rangel-Pineros et al. 2022), VIBRANT (Kieft et al. 2020), VirSorter2 (Guo et al. 2021).

Conclusion

nf-core/viralmetagenome addresses a critical need in viral genomics by providing an automated, scalable solution for untargeted viral genome reconstruction. The pipeline successfully automates the traditionally time-consuming and manual execution process of viral genome assembly reconstruction from short-read metagenomic data through its integrated workflow of *de novo* contig assembly, automated

reference selection, clustering algorithms, and iterative refinement strategies.

Our validation demonstrates the pipeline's broad applicability across diverse results indicate that the pipeline is suitable for a wide range of eukaryotic viral families, achieving it removes the burden of extensive manual curation, and enables consistent, high-quality genome reconstruction while ensuring reproducibility and ease of deployment across different computational environments.

with reproducible results across varied computational settings. As viral surveillance and outbreak response increasingly rely on metagenomic sequencing, automated pipelines like nf-core/viralmetagenome will be essential for the timely identification of pathogen strains. The pipeline represents a significant step forward in making viral genome reconstruction accessible to researchers without requiring extensive bioinformatics expertise, facilitating broader adoption of metagenomic approaches in viral research and public health applications.

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Author Contributions

J.K. designed and implemented the pipeline, performed validation analyses, and wrote the manuscript. P.L. and L. E. K. supervised the project and along with M.B. provided critical feedback. The nf-core community contributed to maintaining the pipeline. All authors reviewed and approved the final manuscript.

Conflict of Interest

The authors declare no competing interests.

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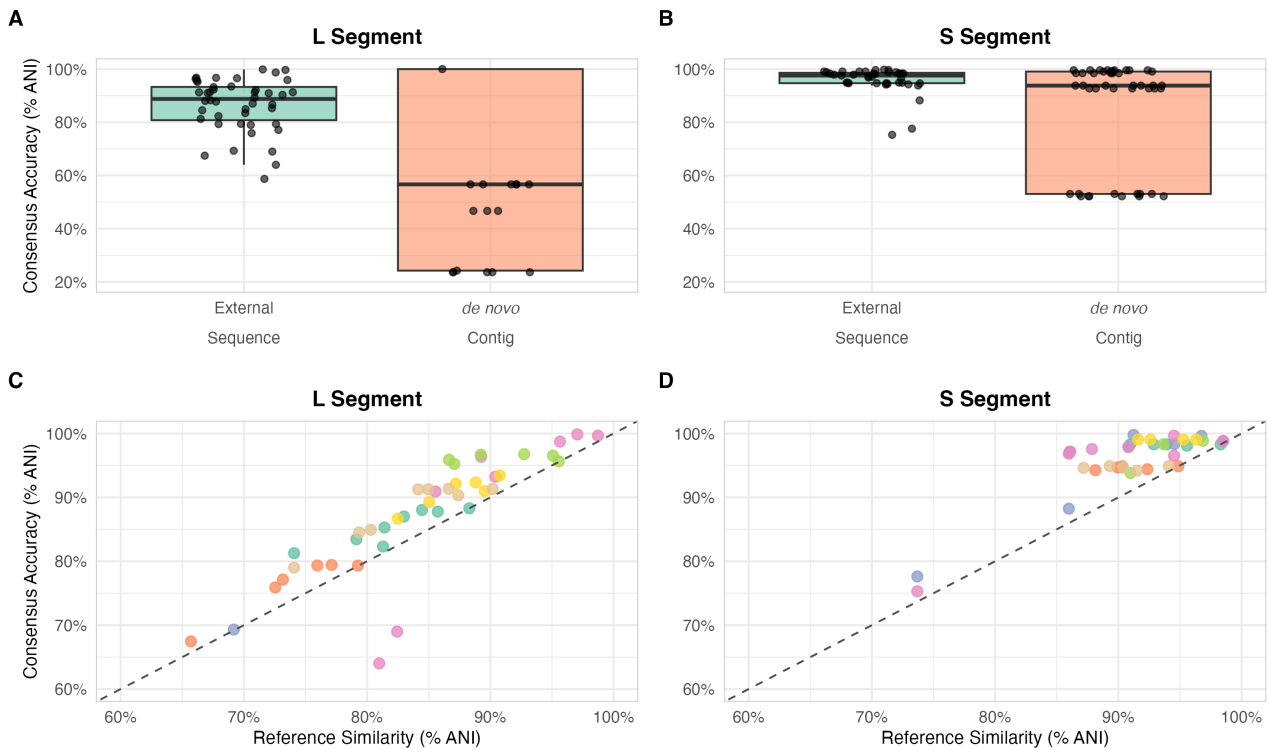


Fig. 2. Impact of scaffolding reference similarity on consensus reconstruction accuracy. The baseline consensus for each sample was generated using the pipeline's automated reference selection from the full unclustered-RVDB (U-RVDB) pool. **(A, B)** Boxplots comparing Consensus Accuracy (% ANI) for the L and S segments. The plots contrast the performance of using an external sequence as the scaffolding backbone (green) against relying solely on the *de novo* contig (orange). For the low-coverage L segment **(A)**, external references significantly improve accuracy, whereas for the high-coverage S segment **(B)**, the *de novo* contig performs comparably to reference scaffolding (sequencing depths detailed in Supplementary Figure S3). **(C, D)** Scatter plots illustrating the relationship between reference similarity (% ANI) and consensus accuracy (% ANI). The dashed line represents the line of identity ($y = x$), and points are colored by sample source (as per Supplementary Figure S2).

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